

WABS

With AI Be Sustainable

AI Agent Specialist · Since 2025

COMPANY OVERVIEW

WABS

WABS specializes in solving real-world business challenges through AI agent technology. We design, develop, and operate AI solutions across logistics, energy, manufacturing, and business automation.

"We see problems first, not technology" — We deeply understand our clients' business challenges and deliver fully working solutions end to end.

Company	**WABS Inc.**
CEO	Hyunjun Choi
Established	2025-06-04
Address	401-166 A-ho, 4F, 79 Gimpohangang 9-ro, Gimpo-si, Gyeonggi-do, Republic of Korea
Email	wabs@wabs.io
Website	https://www.wabs.io/

VISION & MISSION

With AI, Be Sustainable

Contributing to a sustainable future with AI. We connect the potential of technology to real-world industry problems, creating lasting value rather than one-time solutions.

CORE VALUES

Problem-First

Real problem-solving is our reason for being. We start by deeply understanding our clients' business challenges.

Deep Completion

We take full responsibility until every project works in production. We deliver real services, not prototypes.

Big Bowl

We embrace problems beyond the boundaries of domain, technology, and people — connecting logistics, energy, AI, and automation as one.

Sustainable

We pursue long-lasting value, not quick fixes. Our designs consider operations and maintenance from the start.

OUR PEOPLE

WABS is a team of people who believe humanity can build a sustainable future together with AI.

01 Problem Seekers

We ask 'why' before 'how', defining problems in our clients' language.

02 Finishers

We don't back down from challenges — we prove our worth through working results.

03 Boundary Crossers

We go beyond our own expertise, connecting different domains and technologies.

04 Sustainability Advocates

We work with a long-term perspective, driven by the conviction that AI can contribute to humanity's future.

OUR SERVICES

Solving Business Problems with AI

From consulting to development and operations — we define problems and deliver AI solutions.

AI Agent Design & Development

Custom AI agents built on LangGraph and multi-agent architectures to automate your workflows.

LangGraph | Multi-Agent | RAG

Full-Stack SaaS Development

End-to-end SaaS platform development from planning through deployment and operations.

Next.js | Supabase | Docker

Data Visualization & Analytics

Real-time dashboards, interactive 3D visualizations, and BI reports that turn data into insight.

React | Three.js | Recharts

MLOps & GPU Deployment

GPU-based AI inference pipelines deployed automatically through Kubernetes and Triton.

Kubernetes | Triton | Docker

LLM Integration & Agent Consulting

Integrate Claude, GPT, Gemini, and other LLMs into business workflows while optimizing reliability and cost.

Claude API | Prompt Caching | MCP

Synthetic Data & Test Automation

Synthetic data and automated test environments that stay safe and useful inside regulated industries.

LLM-First | Synthetic Data | QA

OUR SOLUTION 1/3

WABSTOA

Our AX framework built on WABS NSWM — it turns a site's equipment, work, and rules into a world model and checks every AI decision against those rules before it runs.

TYPE	PERIOD	DEMO VIDEO
WABS in-house solution	2026.09 - 2026.10	



Architecture

Discovery → Modeling → Integration → Simulation → Pre-check → Revision

CHALLENGE

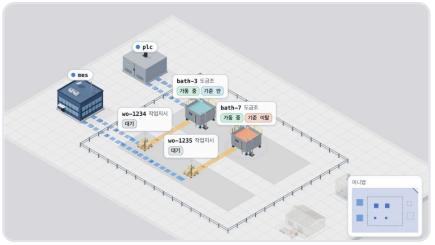
Industrial sites run five to seven separate systems — ERP, MES, LIMS, maintenance — and each one names the same equipment differently. Even a single call such as "can we replenish this chemical now?" meant digging through several systems and shift notes. Adding an LLM agent did not solve it: company rules stayed as sentences inside a prompt, so disallowed actions could not be stopped before they ran and decisions were hard to explain. Departments also used the same words with different meanings, which made it hard to grow one department's system into a company-wide one.

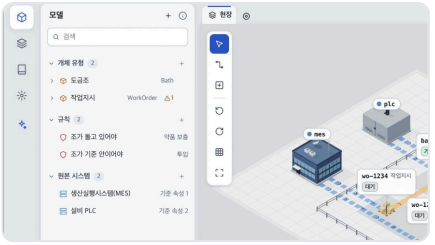
SOLUTION

The site's objects, states, actions, and rules are declared as a world model. It sits on top of existing systems such as MES and PLC without replacing them. The AI (LLM) only interprets what people say and proposes candidate actions; a rule kernel decides before execution whether each one is allowed, denied, or sent to a person for approval, and every denial is recorded with a reason code. All verdicts, approvals, and design decisions go into an append-only ledger, and each department sees the same objects through its own perspective. A visual design studio, a simulator, and an operations console are included. The kernel makes no LLM calls, so the model can be swapped through an OpenAI-compatible adapter.

Key Screens







Key Features

Visual World Modeling

Object types (plating baths, work orders), states, actions, rules, and source systems are defined by drawing them on screen. Designers edit without knowing any file format, and a to-do list points out gaps before saving.

Pre-execution Verdicts

The rule kernel rules on every AI-chosen action before it runs — allow, deny, or ask a person. Blocked actions are explained in plain words and recorded in the ledger with a reason code.

Perspectives on One Source of Truth

Operators, quality, production, and inventory staff each see the same plating bath the way they need it. Every view is projected from one canonical model, so answers never contradict each other, and adding a department leaves existing screens untouched.

Rules That Improve from Records

Verdicts, approvals, and design decisions all accumulate in the ledger. When a rule is revised, WABSTOA measures whether it is better against human judgments and real outcomes, works out which departments must sign off, and only then puts it into operation.

Achievements

5,209
Automated tests

26/26
Spoken requests (demo)

2
Required dependencies

Technology Stack

Python · Pydantic · PyYAML · SQLite · PostgreSQL · FastAPI · Uvicorn · OpenAI-compatible LLM API · Gemma · AWS

OUTCOME

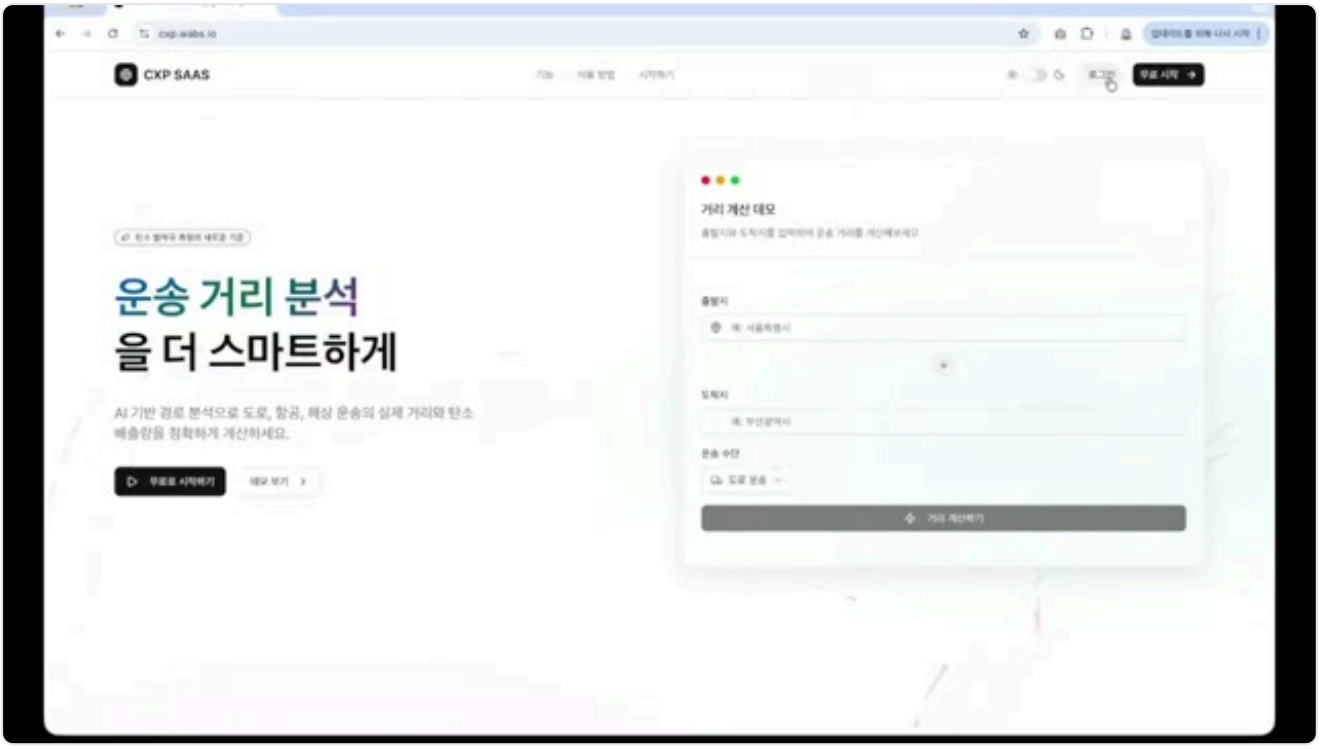
One end-to-end flow now runs on a demo server with an electroplating example. It covers plating baths, work orders, MES, and PLC, from design through simulation, operation, spoken requests, and revision approval. 5,209 automated tests guard kernel principles such as "no state change without a record," and all 26 spoken-request measurements in the demo rehearsal were handled correctly. What the customer organization decided and how it turned out keeps accumulating, and when a rule is revised, WABSTOA measures whether the revision is actually better — against human judgments and real outcomes — and works out who has to approve it.

OUR SOLUTION 2/3

CXP SaaS

LCA transport route analysis platform — from CSV upload to AI route analysis, carbon emission calculation, and visual reporting.

TYPE	PERIOD	TEAM	DEMO VIDEO
WABS in-house solution	2025.12 - 2026.03	2 engineers	



Architecture

CSV Upload → AI Geocoding → Route Analysis → Carbon Calculation → 3D Visualization → Report Export

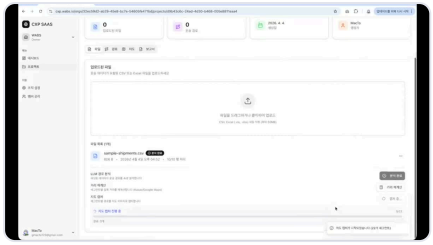
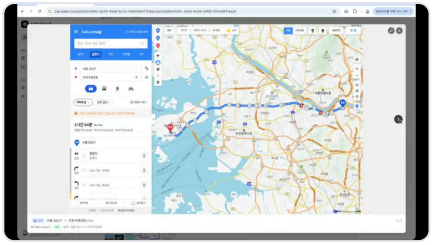
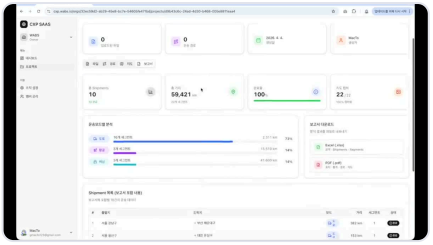
CHALLENGE

Logistics teams and ESG officers calculated per-route carbon emissions by hand, which made data processing slow. Segment-level analysis of multi-modal routes (road, air, sea) was effectively impossible. Applying international emission factors consistently was also a persistent challenge.

SOLUTION

We built an automated geocoding and route-optimization engine powered by Google Gemini 2.5 Flash, packaged as a Next.js 16 + Supabase full-stack SaaS. The pipeline covers bulk CSV upload, AI analysis, interactive 3D globe visualization, and Excel/PDF report export end to end. A multi-tenant architecture provides organization/project-level collaboration with role-based access control.

Key Screens

Key Features

AI Route Analysis

Auto-geocodes origins and destinations with Google Gemini 2.5 Flash and analyzes optimal transport routes.

Multi-modal Decomposition

Automatically decomposes road, air, and sea routes into segments, calculating distance and emissions.

3D Globe Visualization

Visualize transport routes and carbon emissions on an interactive Three.js 3D globe.

Multi-tenant SaaS

Organization/project-level collaboration, role-based access control, and bulk CSV upload.

Achievements

90%+
Analysis time reduction

Road·Air·Sea
Supported transport modes

Excel/PDF
Report formats

Technology Stack

Next.js 16 · React 19 · Tailwind CSS 4 · Supabase · Google Gemini 2.5 Flash · Leaflet · Three.js · AWS EC2 · Docker

OUTCOME

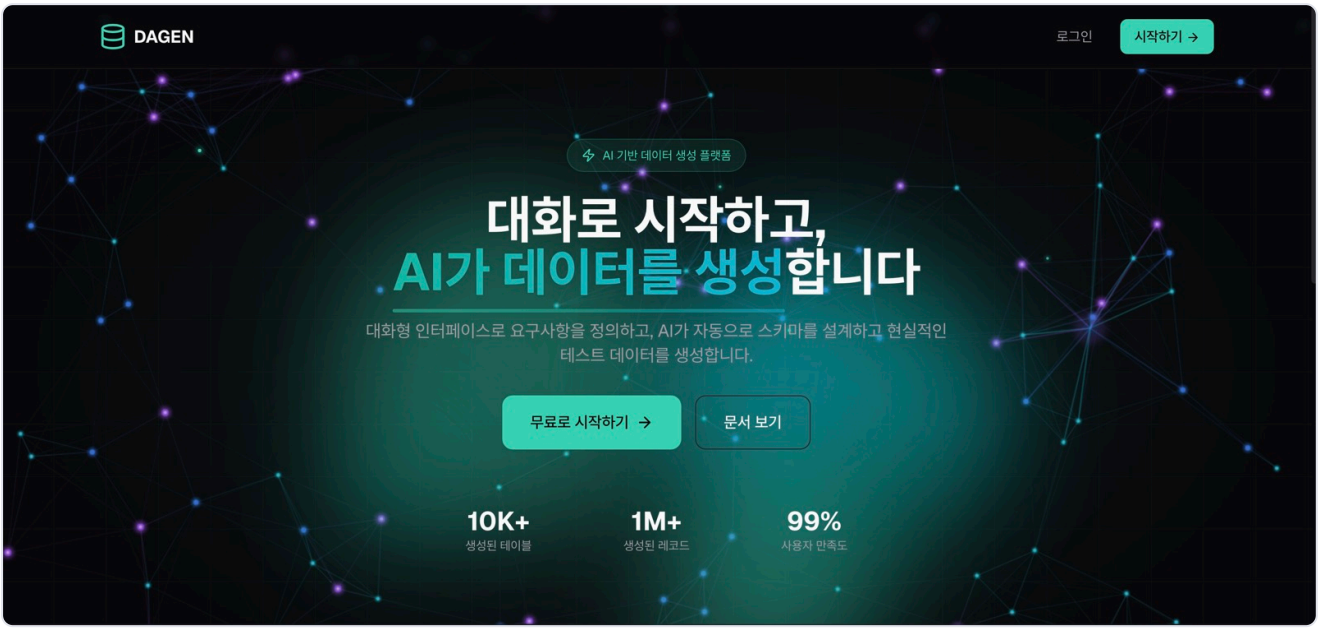
Reduced route-analysis time by over 90% compared to the manual baseline. Multi-modal routes (road, air, sea) are decomposed into segments automatically with distance and emission calculations. International emission factors are applied consistently to deliver accurate, repeatable carbon-footprint data. Live at cxp.wabs.io.

OUR SOLUTION 3/3

DAGEN

LLM-first synthetic financial data platform — generates causally consistent records via persona-based Actor and journey simulation.

TYPE	PERIOD	TEAM	DEMO VIDEO
WABS in-house solution	2025.11 - 2026.04	2 engineers	



Architecture

ERD Upload → Schema Classification → Persona Generation → Journey Simulation →
Consistency Check → Multi-format Export

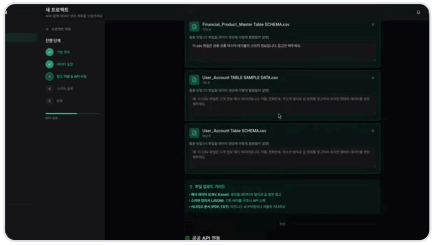
CHALLENGE

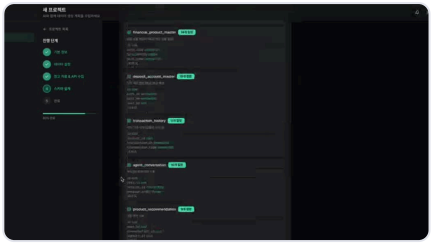
Building financial AI agents demands realistic test data that mirrors actual user behavior, yet regulation makes real financial records off-limits. Conventional random or rule-based synthesizers generate columns independently, so foreign-key integrity, date arithmetic, and monetary constraints cannot all be satisfied at once — leaving teams to patch data manually for every demo and test cycle.

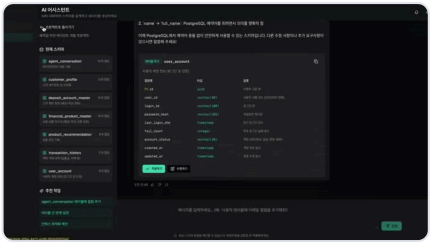
SOLUTION

Built on a Claude API-centric, LLM-first architecture that guarantees row-level consistency. It combines persona-driven Actor generation with journey-based simulation. An LLM-driven schema classifier, event-to-table mapping, and a context pre-generation pipeline resolve business rules and foreign keys automatically. Additional capabilities include automatic ERD image analysis (Vision API), token-cost optimization through Prompt Caching, and SSE-based real-time progress monitoring. Five RDBMS targets (PostgreSQL/MySQL/MSSQL/SQLite/Oracle) and four export formats (CSV/SQL/JSON/Debug Dump) are supported out of the box.

Key Screens







Key Features

LLM-First Synthetic Data

Claude API-centric architecture generates financial synthetic data with row-level causal consistency through Persona and Journey simulation.

Automatic ERD Analysis

Vision API auto-parses ERD images to extract tables, relationships, and constraints, pre-generating context.

5 RDBMS · 4 Export Formats

Supports PostgreSQL/MySQL/MSSQL/SQLite/Oracle and CSV/SQL/JSON/Debug Dump exports.

Real-time Progress Monitoring

SSE-based real-time progress and log streaming with Prompt Caching for token cost optimization.

Achievements

5 engines
Supported RDBMS

4 formats
Export formats

Row-level 100%
Data consistency

Technology Stack

Next.js 14 · Node.js 20 LTS · Express.js · Prisma · PostgreSQL · Redis / Bull Queue · Anthropic Claude API · Google Gemini API · Socket.io · Tailwind CSS

OUTCOME

Generates synthetic data along realistic financial journeys (signup → account opening → deposit → transfer). The data is ready for immediate demo and test use. The LLM consistently enforces date arithmetic such as maturity = open date + term, monetary range constraints, and state-driven logic, removing the overhead of manual data verification. Live as a SaaS at dagen.wabs.io for external use.

REFERENCE PROJECTS

Proven Capabilities

A track record of successfully building and delivering AI solutions across diverse industry domains. Each project is detailed project by project on the following pages.



COMMERCE / MARKETING 2026.02 - 2026.07

LUNA

Microsoft Teams-based multi-agent assistant for a global consumer-electronics company's D2C subsidiaries — supports marketers from coupon policy questions to hands-on Magento, AEM and Adobe Target setup in one conversation.



COMMERCE / MARKETING 2025.07 - 2025.11

LAMDA

Desktop AI agent for LG CNS in-house marketing analytics — automates BigQuery/GA4 data analysis and site-UX inspection from natural-language queries.



AI / DEVOPS 2025.09 - 2026.01

Harness Monitor

AI skill management dashboard — search, visualization, automated configuration, and orchestration in one place.



ENERGY / MANUFACTURING 2025.06 - 2025.12

ENMS

Smart-factory energy monitoring and 3D visualization platform — real-time power surveillance, carbon emissions analytics, and metaverse integration.



MULTI-AGENT OPTIMIZATION 2025.03 - 2026.10

Multi-AI Agent System

Enterprise multi-agent assistant that handles groupware work through conversation — 10 domain agents and 45 tools, extended with personalized recommendations and workflow automation.



BUSINESS AUTOMATION 2025.01 - 2025.06

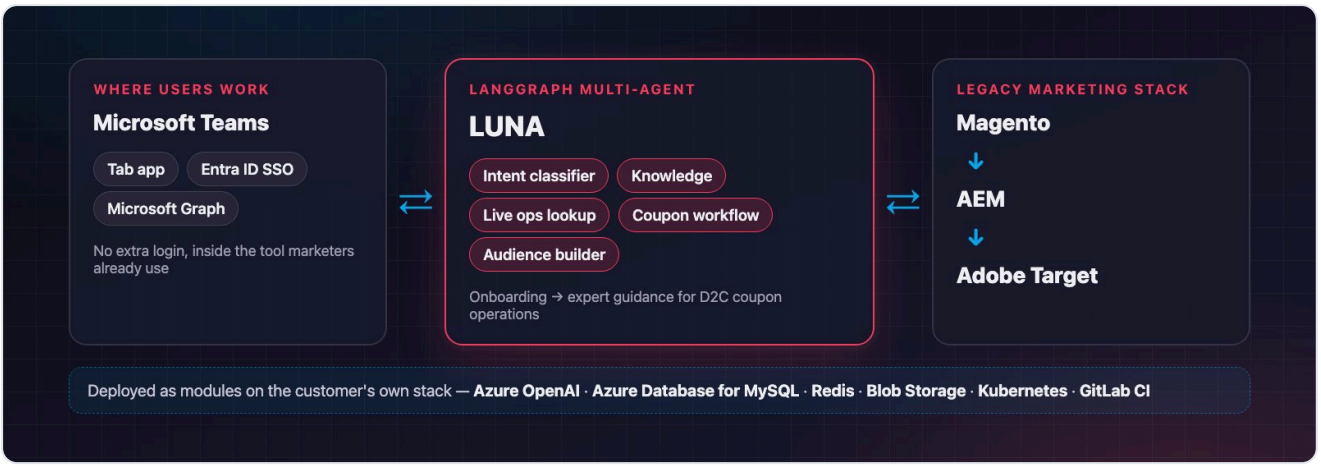
Meeting Assistant

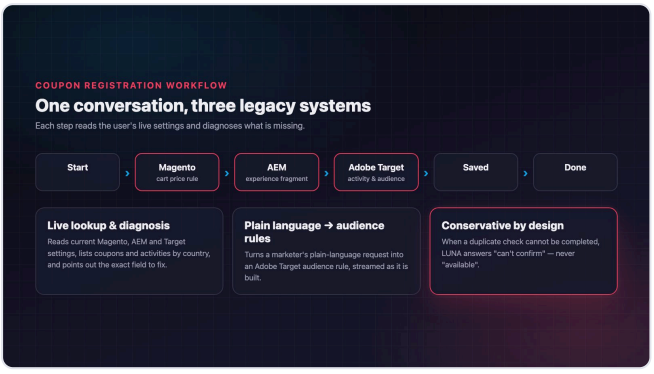
GPU-powered AI meeting-minutes automation — audio upload, speaker diarization, STT, AI summarization, and email delivery.

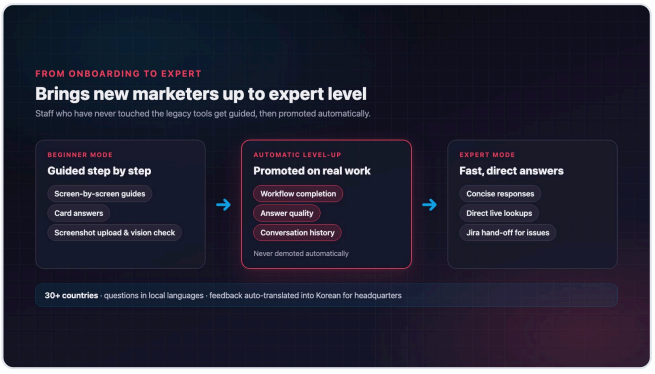
COMMERCE / MARKETING 2026.02 - 2026.07

LUNA

Microsoft Teams-based multi-agent assistant for a global consumer-electronics company's D2C subsidiaries — supports marketers from coupon policy questions to hands-on Magento, AEM and Adobe Target setup in one conversation.







CHALLENGE

Overseas marketers at a global consumer-electronics company juggled three legacy systems — Magento, AEM and Adobe Target — for every coupon.

SOLUTION

We led the design of a LangGraph multi-agent assistant that runs as a tab inside Microsoft Teams, and took part in building it.

OUTCOME

Now in production as the Teams-based work assistant for a global consumer-electronics company's D2C subsidiaries across 30+ countries.

30+ Countries supported **4 systems** Legacy systems integrated **5 agents** Collaborating agents

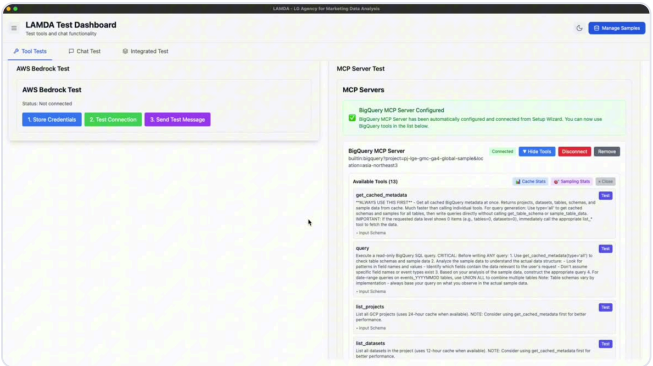
Microsoft Teams (Tab App) · Entra ID SSO · Azure OpenAI · LangGraph · FastAPI · React 18 + TypeScript · Adobe Target API · Playwright · Azure Database for MySQL · Azure Blob Storage · Kubernetes · GitLab CI

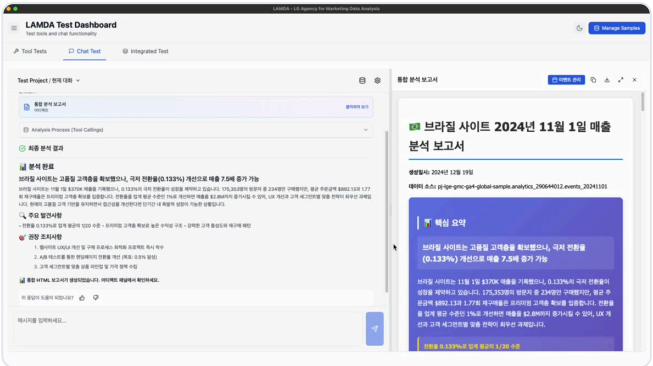
COMMERCE / MARKETING 2025.07 - 2025.11

LAMDA

Desktop AI agent for LG CNS in-house marketing analytics — automates BigQuery/GA4 data analysis and site-UX inspection from natural-language queries.







CHALLENGE

The LG CNS marketing team needed to analyze GA4 data in BigQuery alongside live website UX.

SOLUTION

We anchored the analytics engine on AWS Bedrock Claude Sonnet 4 and integrated a BigQuery MCP server.

OUTCOME

Delivered to LG CNS as an in-house tool that covers the full analytics flow.

5 agents Specialist agents **8 formats** Artifact formats **Up to 10 rounds** Multi-turn tool calls

Tauri 2.x (Rust) | React 18 + TypeScript | AWS Bedrock (Claude Sonnet 4) | Google BigQuery | GA4 API | Model Context Protocol (MCP) | Playwright | SQLite | Vite | Chart.js / Plotly

DEMO VIDEO

Scan the QR to watch the actual solution in action.

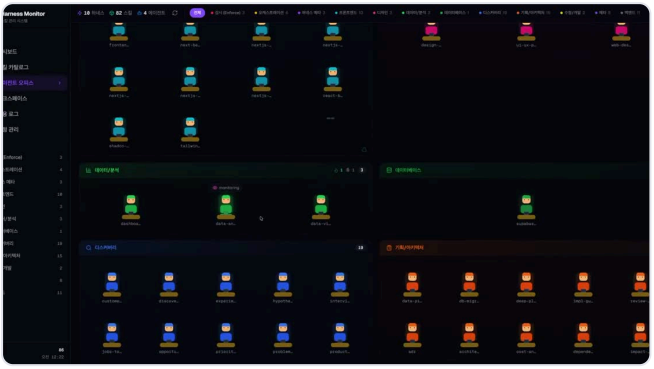
`https://www.wabs.io/portfolio/videos/lamda.mp4`

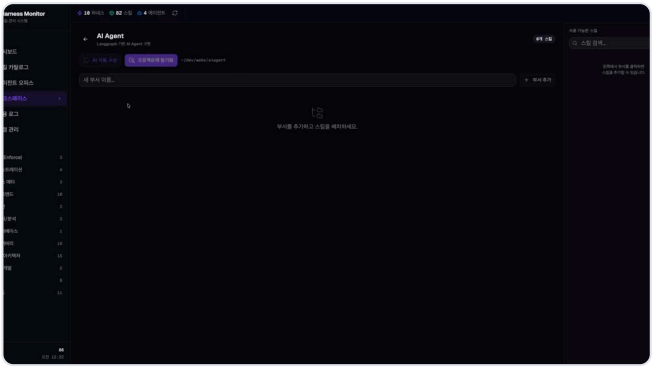
AI / DEVOPS 2025.09 - 2026.01

Harness Monitor

AI skill management dashboard — search, visualization, automated configuration, and orchestration in one place.







CHALLENGE

Managing 62+ AI skills systematically and matching the optimal combination to each project was inefficient by hand.

SOLUTION

We developed a natural-language search and automated skill-composition system powered by Gemini 2.5 Flash.

OUTCOME

Centralized management of 62+ AI skills with natural-language search.

62+ Skills managed **3 stages** Auto-configuration **4-layer** Architecture

Next.js | TypeScript | Prisma | Gemini 2.5 Flash | Docker

DEMO VIDEO

Scan the QR to watch the actual solution in action.

https://www.wabs.io/portfolio/videos/harness-monitor.mp4

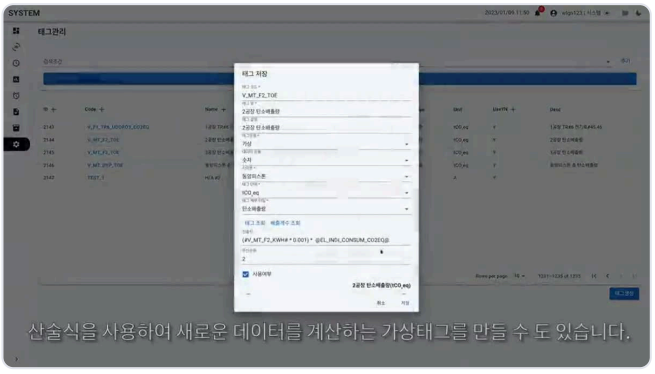
ENERGY / MANUFACTURING 2025.06 - 2025.12

ENMS

Smart-factory energy monitoring and 3D visualization platform — real-time power surveillance, carbon emissions analytics, and metaverse integration.







CHALLENGE

Smart factories lacked a system to monitor and manage energy consumption in real time, so energy waste could not be detected proactively.

SOLUTION

We built an IoT-sensor-based real-time energy data pipeline.

OUTCOME

Real-time monitoring of plant-wide energy consumption enables instant detection of abnormal patterns.

24/7 Real-time monitoring **IoT sensors** Data ingestion **3D metaverse** Visualization

React TypeScript Recharts Spring Boot InfluxDB MariaDB 3D Metaverse

DEMO VIDEO

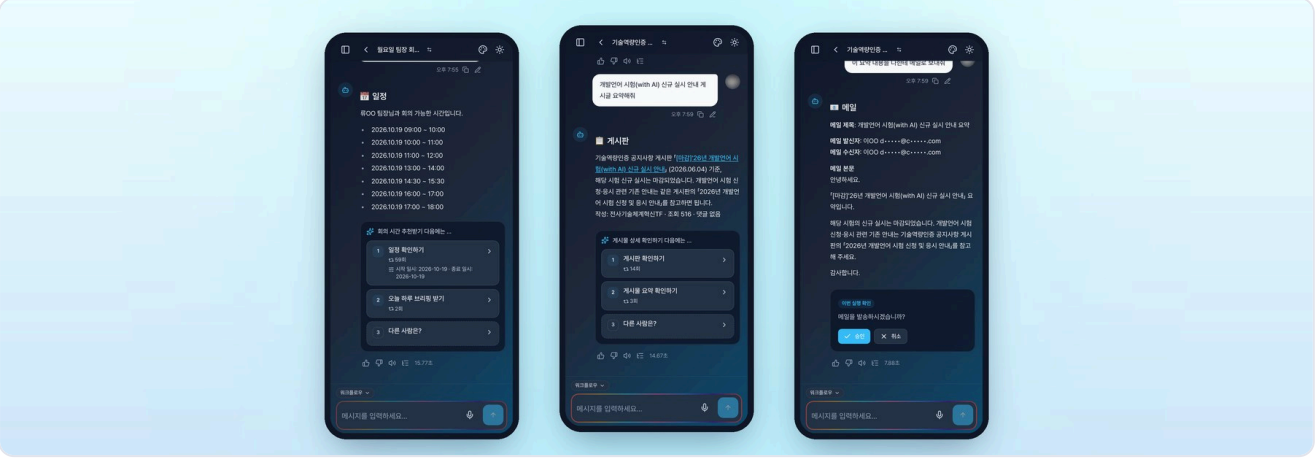
Scan the QR to watch the actual solution in action.

`https://www.wabs.io/portfolio/videos/enms.mp4`

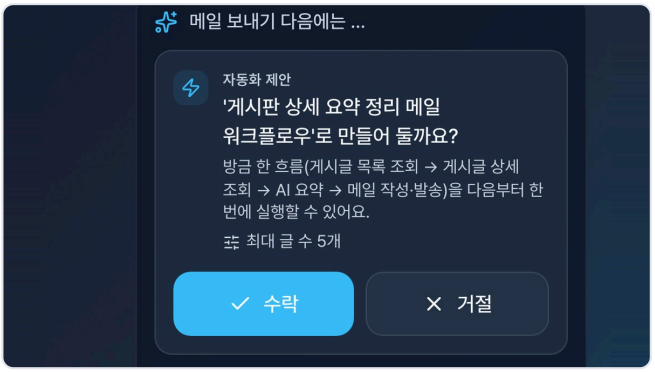
MULTI-AGENT OPTIMIZATION 2025.03 - 2026.10

Multi-AI Agent System

Enterprise multi-agent assistant that handles groupware work through conversation — 10 domain agents and 45 tools, extended with personalized recommendations and workflow automation.







CHALLENGE

Mail, calendar, approvals, and directory lived in separate menus, so booking one meeting meant manual lookups, free-time checks, and invites.

SOLUTION

A three-stage LangGraph pipeline routes each request to 10 domain agents, paired with learned next-action recommendations and a workflow agent.

OUTCOME

Users query and act on six groupware areas in a single sentence, while recurring work runs on schedule as a personalized assistant.

10 agents Domain agents **45 tools** Business tools **5 + custom** Built-in workflows

LangGraph · LangChain · deepagents · Azure OpenAI · FastAPI (SSE) · PostgreSQL + pgvector · React 19 · Azure Speech · Langfuse

DEMO VIDEO

Scan the QR to watch the actual solution in action.

https://www.wabs.io/portfolio/videos/multi-ai-agent.mp4

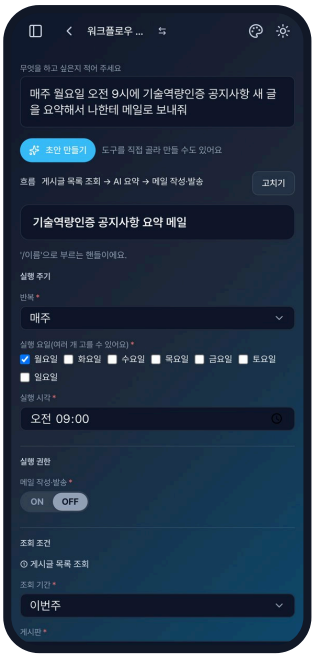
Architecture

Intent → Routing → Delegation → HITL approval → Next actions → Workflow offer

Key Screens



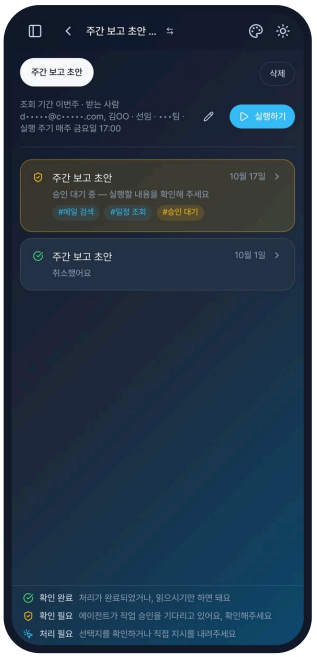
1 Pick a template — five built-ins or build your own



2 Describe it in words — a draft with flow and slots filled in



3 Schedule & permissions — e.g. Fridays 17:00, per-step approval



4 Track runs — pending approvals and results at a glance

Key Features

Next-action recommendations

Up to three cards suggest what to do next after each answer. Markov transition statistics blend personal and company-wide history, growing more personal as usage accumulates, and a tap opens a pre-filled form ready to run.

Workflow agent

Bundles multi-step flows such as lookup → AI summary → send mail and runs them with one tap, a chat /command, or on a daily, weekly, or monthly schedule. Ships with five built-ins, including board digests and attendance checks.

Proactive automation offers

When a user repeats the same flow on the same weekday or on three of the last 14 days, the system offers to save it as a workflow. Detection runs on code rules, not the LLM, so results stay consistent.

Custom workflows from plain language

Describe the task in words and the agent drafts a flow plus a single input form from 45 tools. Earlier step outputs auto-fill later fields, and only flows passing 12 validation rules are saved.

Human-in-the-loop approval

Seven write actions — sending mail, booking meetings, approving documents — pause the graph for an approval card. State is checkpointed so runs resume minutes later, and each workflow step has its own approval switch.

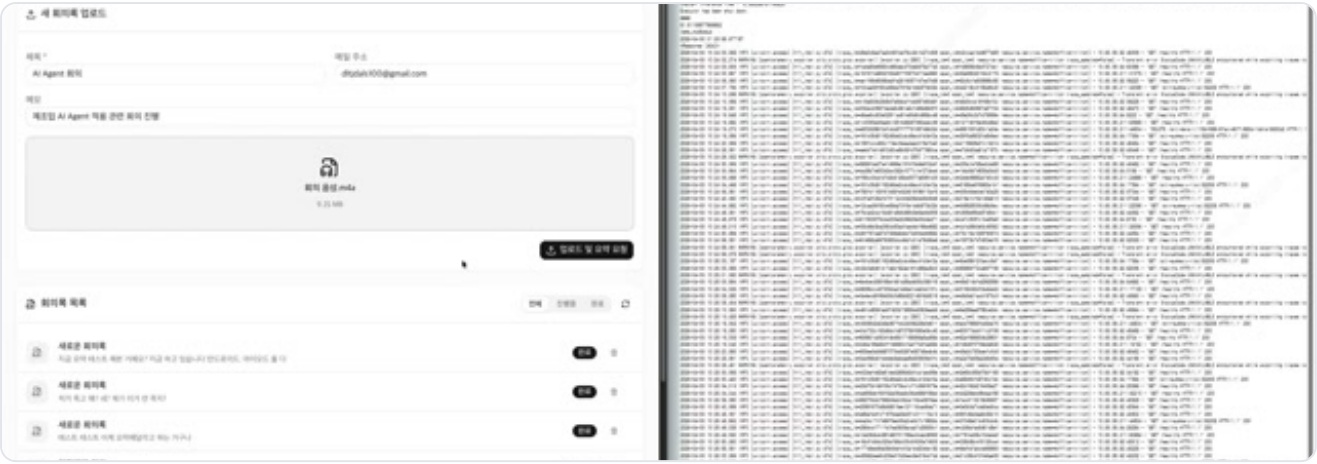
Cold start & privacy

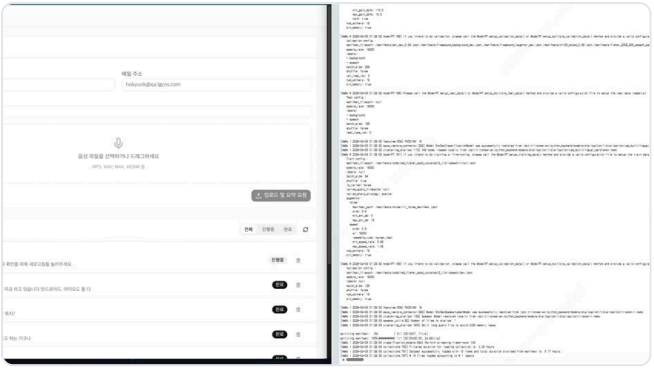
New users start from a synthetic seed of 90 virtual users over 28 days (17,977 events). Behavior logs never store mail subjects or bodies, and users can reset their own learning data at any time.

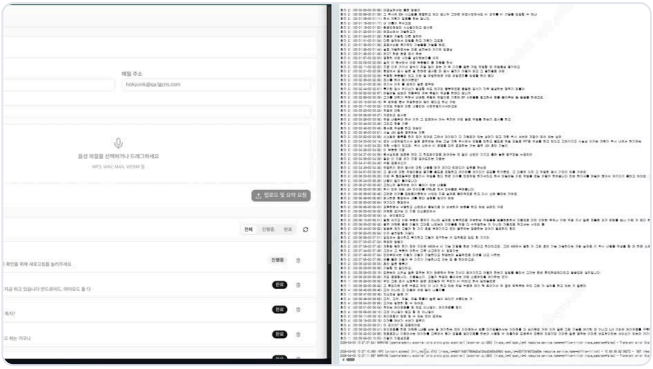
BUSINESS AUTOMATION 2025.01 - 2025.06

Meeting Assistant

GPU-powered AI meeting-minutes automation — audio upload, speaker diarization, STT, AI summarization, and email delivery.







CHALLENGE

Drafting meeting minutes consumed excessive time, and low speaker-identification accuracy forced manual transcription.

SOLUTION

We built a six-stage AI pipeline from audio upload to email delivery.

OUTCOME

Fully automated the path from raw recording to structured minutes.

6-stage AI Pipeline **Private/Hybrid/Cloud** Deployment options **Triton Server** GPU acceleration

FastAPI | NVIDIA Triton | NeMo Diarization | Whisper v3 | GPT-4.1 | Kubernetes | RabbitMQ | MinIO | MongoDB | Grafana

DEMO VIDEO

Scan the QR to watch the actual solution in action.

`https://www.wabs.io/portfolio/videos/meeting-assistant.mp4`

Overall Capabilities

9	7	64+	Full-stack AI
Projects delivered	Domains covered	Technologies	Team expertise

Technology Stack

Frontend	Next.js · React · Tailwind CSS · Leaflet · Three.js · TypeScript · Tauri 2.x (Rust) · Vite · Chart.js · Plotly · Recharts · 3D Metaverse
Backend & Database	Python · Pydantic · PyYAML · SQLite · PostgreSQL · FastAPI · Uvicorn · Supabase · Node.js · Express.js · Prisma · Redis · Bull Queue · Socket.io · Azure Database for MySQL · Google BigQuery · Spring Boot · InfluxDB · MariaDB · RabbitMQ · MinIO · MongoDB
AI & ML	OpenAI-compatible LLM API · Gemma · Google Gemini 2.5 Flash · Anthropic Claude API · Google Gemini API · Azure OpenAI · LangGraph · AWS Bedrock (Claude Sonnet 4) · Model Context Protocol (MCP) · LangChain · deepagents · pgvector · Azure Speech · Langfuse · NVIDIA Triton · NeMo Diarization · Whisper · GPT-4.1
Infra & DevOps	AWS · AWS EC2 · Docker · Playwright · Azure Blob Storage · Kubernetes · GitLab CI · Grafana
Enterprise Integration	Microsoft Teams (Tab App) · Entra ID SSO · Adobe Target API · GA4 API

Milestones

2025.06	WABS incorporated — AI agent specialists
2025.06–12	Built ENMS, an energy monitoring system for smart factories
2025.07–11	Built and delivered LAMDA, a marketing-analytics AI agent, for LG CNS
2025.12–	Developed our CXP SaaS platform and launched cxp.wabs.io
2026.02–07	Led the design of LUNA, a Teams-based agent for a global consumer-electronics company's D2C subsidiaries (in production)
2026.04	Launched DAGEN, our synthetic financial data platform (dagen.wabs.io)
2026.09–	Developing WABSTOA, our AX framework that checks AI actions against a world model before they run (demo stage)
2026.10	9 projects across 7 domains, 64+ technologies



WABS

With AI Be Sustainable

EMAIL

wabs@wabs.io

WEBSITE

(main) https://www.wabs.io/

(reference) https://www.wabs.io/portfolio

ADDRESS

401-166 A-ho, 4F, 79 Gimpohangang 9-ro,
Gimpo-si, Gyeonggi-do, Republic of Korea

Build the Future Together

Have a problem you'd like to solve with AI?
Let's build a sustainable solution together.

Get in Touch